NeuroSonic: a laptop-assisted, offline-first smartphone prototype for spoken walking-path awareness and navigation for visually impaired pedestrians

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Research prototype report. All results measured on one laptop on 2026-11-02/03 unless stated otherwise.

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Live demo video: https://drive.google.com/file/d/1yJS2f8_ee3ZwV55mkA9e1c6mm9f5xkYd/view?usp=sharing

Abstract. We describe NeuroSonic, a research prototype in which an iPhone acts as the camera, microphone, GPS receiver and speaker of a perception and navigation system that runs on a laptop over local Wi-Fi. The system grew out of WOAD, a near-infrared and depth obstacle detector that needs a sensor the phone does not have. For the phone's RGB camera we benchmarked four detectors under identical conditions on 365 near-infrared validation images containing 129 people. RF-DETR Nano gave the highest person AP@0.5 (0.660; SSDLite 0.542, YOLO26n 0.619, YOLO26s 0.628) and the highest recall at the operating threshold (0.729), with 21.3 ms mean GPU latency on an RTX 4050 Laptop GPU. It runs as the primary detector with an automatic fall-back to SSDLite on the CPU. Detections are tracked with a ByteTrack-style association that compensates for camera motion estimated with pyramidal Lucas–Kanade optical flow and RANSAC. On top of the tracks, two conservative events, "entering the walking path" and "crossing in front", are spoken with an offline neural voice. Offline speech commands, an offline OpenStreetMap router and an optional, opt-in cloud scene description complete the system. In loopback browser tests with a simulated camera, the pipeline sustained about 5 frames/s with a median capture-to-answer time of 67 ms. Three engineering problems were found and corrected: idle PyTorch CPU threads (712 % → 121 % of one core), per-request inference threads, and the GPU power state at low frame rates (partly mitigated with FP16, with unchanged person AP@0.5). A 10-minute soak test ran without errors. NeuroSonic has *not* been validated with an iPhone in real streets or with visually impaired users. It is not a safety device. We publish a validation protocol for these tests.

Keywords: assistive technology; visual impairment; real-time object detection; RF-DETR; multi-object tracking; camera-motion compensation; offline speech; pedestrian navigation; edge computing.

1. Introduction

A pedestrian with a visual impairment needs two kinds of information that a white cane and a map application do not give together: *what is in front of me now* (people stepping into the way, a bicycle crossing, a chair in a corridor) and *where do I turn*. Smartphones carry a camera, a microphone, GPS and a

speaker. Running modern detectors at several frames per second on the phone itself is difficult in a browser application, and cloud services need a reliable internet connection and send images off the device.

NeuroSonic explores a middle way. The phone is a thin client in Safari. A laptop with a consumer GPU on the same Wi-Fi does the perception, speech recognition, speech synthesis and routing *without internet access*. The project started from WOAD (Wearable Obstacle Avoidance Device) [1], a two-stream near-infrared and depth model. WOAD requires a CS30 time-of-flight sensor, which no phone has, so the phone camera path had to be built separately. This paper reports how that path was designed and measured, and what remains untested.

The contributions are:

1. a working end-to-end prototype (phone client, laptop server, 28 HTTP endpoints) with offline walking guidance, event alerts, voice commands, neural speech and offline routing;
2. a controlled comparison of four real-time detectors on the same images, leading to RF-DETR Nano with an SSDLite fall-back;
3. a camera-motion-compensated tracker and two conservative, rule-based temporal events designed to avoid false alarms from the user's own movement;
4. a measured account of three performance problems in GPU serving at a low, fixed frame rate, with before/after data;
5. an honest separation of what was measured on the laptop, what was simulated in a browser, and what still needs an iPhone and field tests, with a protocol for the latter.

2. Problem statement and objectives

Problem. Provide short, timely spoken information about recognised objects in the walking path and route guidance to a pedestrian holding or wearing a phone, without depending on internet access, without storing camera images, and without overstating what the system can see.

Objectives.

1. O1: process phone camera frames at ≈5 frames/s with an answer well within one frame interval (200 ms) on the laptop;
2. O2: choose the detector by measured accuracy, latency and licence, and keep a CPU fall-back;
3. O3: announce movement (entering, crossing) without announcing the user's own walking as motion;
4. O4: keep all local functions working offline; make cloud description opt-in;
5. O5: never state distances or safety; say "appears clear", not "safe";
6. O6: verify each function with automated tests, and state plainly what is not verified.

3. Related work

Predecessor. WOAD [1] fuses near-infrared and colour-mapped depth streams with transformer-based cross-modal attention in a YOLO-style detector. In earlier work in this project, we measured its cross-dataset person AP@0.5 at 0.17–0.22. We could not reproduce the authors' reported 0.669 mAP, and the model needs its sensor, so in NeuroSonic it is limited to recorded data.

Real-time detection. Single-stage detectors such as YOLO [2] and SSD [3] made real-time detection practical. SSDLite with a MobileNetV3 backbone [4] targets mobile CPUs. DETR [5] introduced set prediction with transformers and removed non-maximum suppression. RF-DETR [6] is a real-time DETR variant from Roboflow built on a DINOv2 backbone [7] and released under Apache-2.0. YOLO26 [8] is the current Ultralytics family (AGPL-3.0). All detectors used here were trained on COCO [9].

Tracking. SORT [10] associates detections to Kalman-predicted tracks by IoU. ByteTrack [11] also uses low-confidence detections in a second association stage, which keeps tracks alive through occlusion. Our tracker follows ByteTrack's two-stage idea. Instead of a Kalman filter, it uses a camera-motion-compensated prediction.

Camera motion. Sparse optical flow with Shi–Tomasi corners [12] and the Lucas–Kanade method [13] in its pyramidal form [14], followed by a RANSAC [15] fit of a similarity transform, is a standard way to estimate global image motion. We use the OpenCV implementations [16].

Speech and maps. Offline recognition uses Vosk, built on Kaldi [17]. Speech synthesis uses Piper [18], which is based on VITS [19]. Routing uses A* [20] on an OpenStreetMap [21] extract.

We did not carry out a systematic review of assistive-navigation literature for this manuscript. The comparison in this paper is between components we ran ourselves.

4. System design

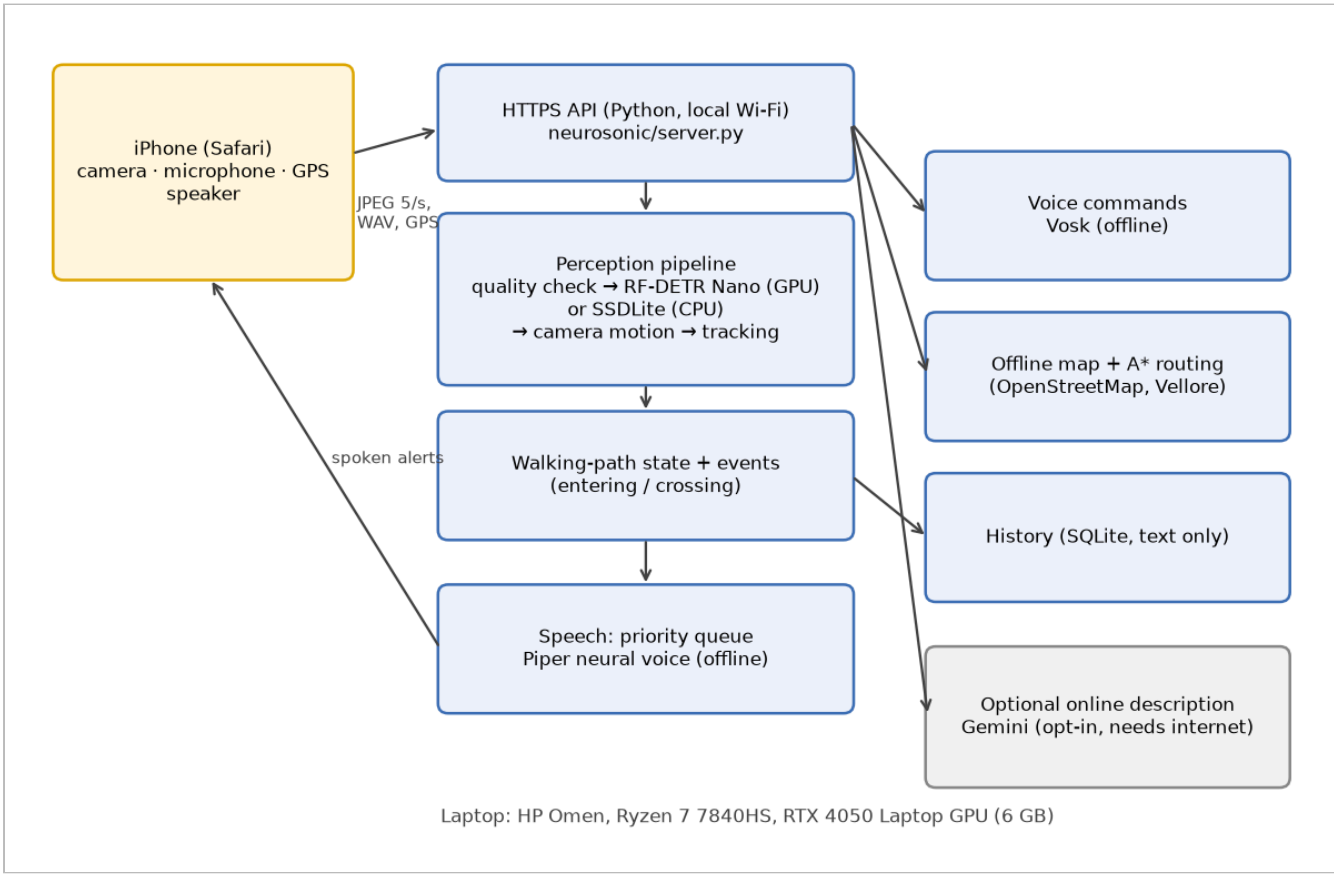


Figure 1. Deployment. The phone is a browser client. All local engines run on the laptop. Only the optional scene description uses the internet, and only after the user opts in.

The server is a standard-library Python HTTP(S) server. The phone loads a web application (HTML/JavaScript) from it and then:

- posts a 320-pixel-wide JPEG every 200 ms (400 ms if the laptop reports the CPU detector), with one request in flight;
- sends recorded speech as WAV for command recognition;
- uses GPS locally, sending positions only to request a route;
- plays spoken messages through a priority queue (urgent > system > obstacle > navigation/scene > info) with cooldowns and interruption.

HTTPS uses a locally generated certificate authority, so that Safari grants camera and microphone access. History is stored in SQLite as text only. No camera image is written to disk or to logs.

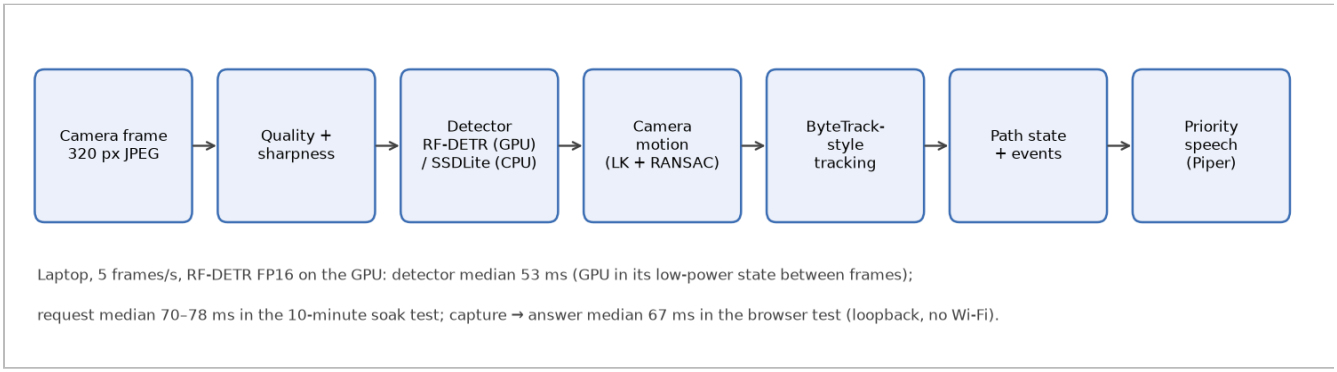


Figure 2. Per-frame perception pipeline on the laptop.

5. Methods

5.1 Detector selection

Four candidates were run through the same harness (`tools/benchmark_detectors.py`):

- the previous production detector, SSDLite320-MobileNetV3-Large (torchvision);
- RF-DETR Nano (rfdetr 1.11.2);
- YOLO26n and YOLO26s (ultralytics 8.4.198, benchmark only).

Quality was measured as person average precision with the project's own AP code, on 365 near-infrared validation images containing 129 annotated people, from the WOAD dataset. These are the only locally available images with person boxes. They are not phone RGB images, so they rank the detectors but do not predict street performance. We also report precision and recall at the production confidence threshold of 0.40.

Latency was the mean over 200 (GPU) or 60 (CPU) recorded 640×480 frames, back to back, FP32, batch 1, after 10 warm-up frames.

Licence was a selection criterion, because the system is a network service: AGPL-3.0 would oblige release of the source to users.

5.2 Detector runtime

RF-DETR runs on CUDA from local weights. Downloads are blocked at run time. Its detections are restricted to 23 obstacle-relevant COCO classes at confidence ≥ 0.40. A wrapper falls back to SSDLite on the CPU in these cases:

- CUDA, the package or the weights are missing;
- warm-up fails;
- a run-time inference fails, in which case the same frame is re-analysed by SSDLite.

Three serving changes were made after measurement (section 7.3):

1. PyTorch is limited to two CPU threads while the GPU detector is active;
2. all inference runs on a single long-lived worker thread rather than on the HTTP server's per-request threads;
3. RF-DETR runs in FP16 (half precision) without graph compilation, with FP32 as fall-back.

5.3 Camera motion and tracking

Camera motion. Up to 150 Shi–Tomasi corners are taken outside the detected boxes and followed into the next frame with pyramidal Lucas–Kanade flow. A similarity transform (shift, scale, rotation) is fitted with RANSAC. If there are fewer than 12 corners or inliers, the camera motion is "unknown".

Tracking. Each track's box is predicted by applying the camera transform to its last box.

- Detections with confidence ≥ 0.55 are matched first, greedily by IoU, within each class.
- Detections between 0.40 and 0.55 can only extend existing tracks, as in ByteTrack.
- Tracks end after three missed frames.
- Every update stores the object's displacement *relative to the background*: the observed motion minus the camera-predicted motion.

5.4 Walking-path state and events

Walking path. The walking path is a lower-central trapezoid of the image. Its state is one of CLEAR_VIEW, POSSIBLE_OBSTACLE, BLOCKED_VIEW, UNCERTAIN (dark or blurred frame) or CAMERA_UNAVAILABLE. A state is confirmed by 2-of-3 or 3-of-3 consecutive frames before it is spoken.

Event preconditions. Events are only considered for confirmed tracks: at least three updates over at least 0.4 s, of people, vehicles or animals.

Event rules.

- *Entering:* the object was outside the path at the start of a 1.5 s window, overlaps it now, and moves towards it at ≥ 0.12 image widths per second relative to the background.
- *Crossing:* the object's centre passes from below 35 % to above 65 % of the width (or back) within 3 s, at ≥ 0.15 widths/s, with the bottom of its box in the lower half of the image.

Suppression.

- Nothing is announced while the camera shifts more than 6 % of the width per frame, zooms more than 4 %, or has no motion estimate.

- Nothing is announced in BLOCKED_VIEW, UNCERTAIN or CAMERA_UNAVAILABLE.
- Each track triggers an event at most once.
- Events are at least 3 s apart.

"Approaching" was deliberately not implemented. With a moving camera, box growth is dominated by the user's own walking. No message states a distance or that the way is safe.

5.5 Speech, commands, routing and optional cloud description

- **Speech output:** Piper with an Indian-English voice synthesises speech on the laptop. Results are cached, and the phone's own voice is the fall-back.
- **Voice commands:** Vosk recognises commands in two passes (grammar-constrained, then free), combined by an intent parser.
- **Routing:** A* with a haversine heuristic on an OpenStreetMap extract of Vellore (31,439 path nodes, 813 places). Turn instructions are spoken about 70 m and 25 m ahead, and the route is recalculated after two off-route fixes.
- **Cloud description:** an optional scene description sends one photo to Gemini, only after explicit consent. It is rate-limited and has timeouts. Its output is filtered so that it never states distances or safety.

6. Experimental setup

Table 1. Test environment.

Item	Value
Laptop	HP Omen; AMD Ryzen 7 7840HS (8 cores / 16 threads); 15.3 GB RAM; NVIDIA GeForce RTX 4050 Laptop GPU, 6 GB; driver 592.82
Software	Windows 11; Python 3.12.10; PyTorch 2.15.0+cu132; torchvision 0.29.1; rfdetr 1.11.2; OpenCV; Piper 1.8.0; Vosk
Environments	`.venv` (CPU-only, original); `.venv-gpu` (CUDA, used for the GPU results)
Detector quality data	365 near-infrared validation images, 129 people (WOAD dataset)
Live-pipeline input	a 640×480 clip that composites a real recorded person riding a bicycle over a real recorded empty corridor (`tools/make_event_clip.py`), played as Chrome's fake camera, or streamed by `tools/soak_test.py`
Browser tests	headless Chrome, iPhone-sized viewport, fake camera and microphone, DNS blocked; server on loopback

No measurement in this paper includes Wi-Fi transmission, iPhone JPEG encoding, real street scenes or users.

7. Results

7.1 Detector comparison

Table 2. Detector comparison (Phase 1 benchmark, `outputs/benchmarks/phase1/results.json`). Latency: mean per 640×480 frame, FP32, batch 1. Quality: person, 365 near-IR images.

Detector	GPU ms	CPU ms	AP@0.5	AP@0.5:0.95	Precision @0.40	Recall @0.40	Licence
SSDLite320 (previous)	42.3	43.7	0.542	0.208	0.607	0.574	BSD-3
RF-DETR Nano (selected)	21.3	121.9	**0.660**	**0.304**	0.588	**0.729**	Apache-2.0
YOLO26n	12.4	27.8	0.619	0.268	0.633	0.628	AGPL-3.0
YOLO26s	11.8	57.5	0.628	0.292	0.600	0.651	AGPL-3.0

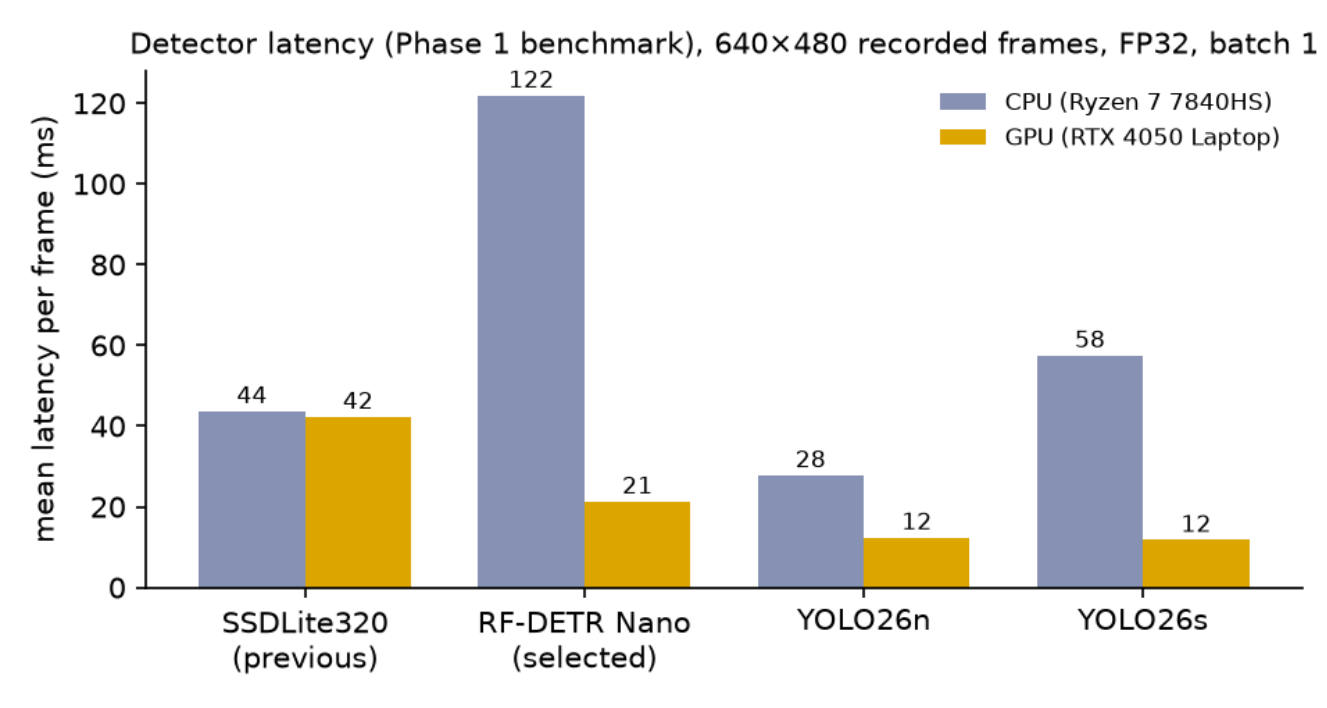


Figure 3. Detector latency.

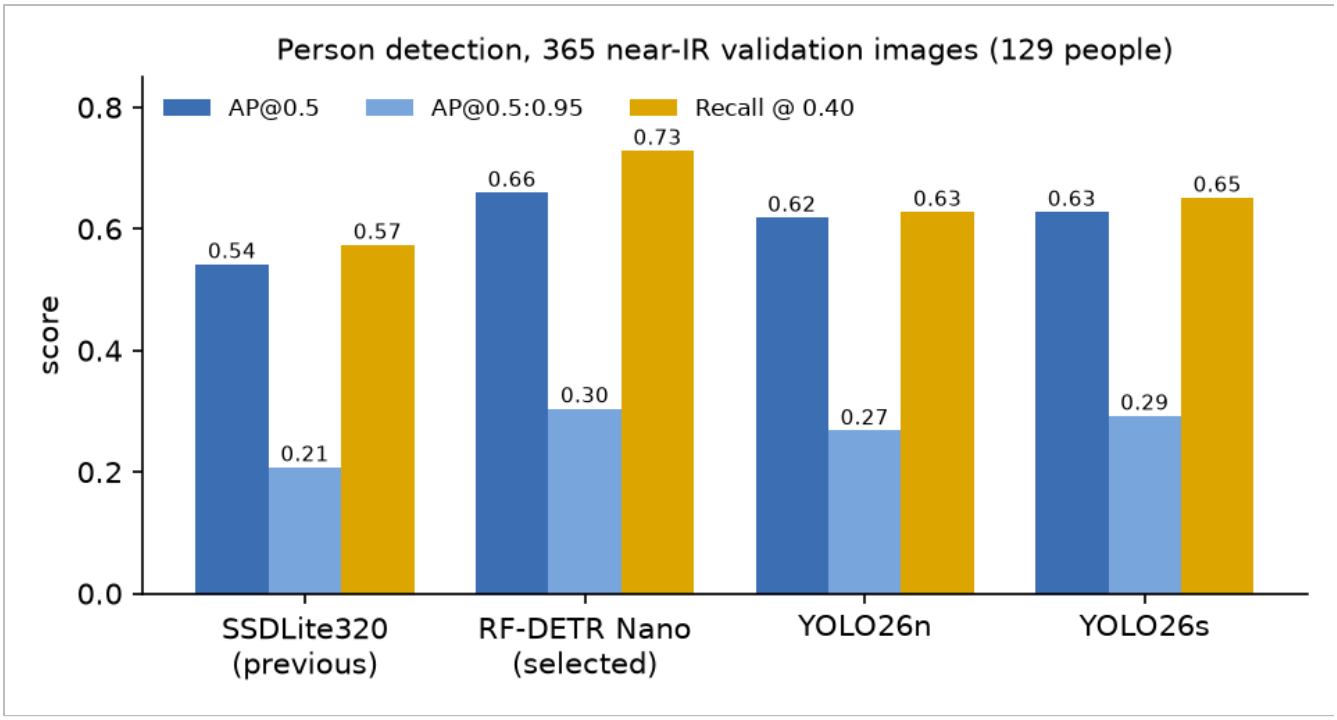


Figure 4. Person detection quality on the near-IR validation set.

Choice of RF-DETR Nano. It had the best AP and recall. Recall matters most for a warning system, because a missed person is worse than an extra caution. Its 21 ms GPU latency fits a 200 ms frame interval, and its licence is permissive.

Trade-offs. The YOLO26 models are faster, but they rank lower on recall and are AGPL-3.0. RF-DETR is too slow on the CPU (122 ms), so SSDLite remains the fall-back. RF-DETR's precision at 0.40 (0.588) is slightly lower than the others', so more false detections are expected. The temporal confirmation in sections 5.3–5.4 is meant to absorb them.

7.2 FP16 inference

RF-DETR was run in FP32 and in FP16 (half precision, no graph compilation) with the same weights and code path (`tools/benchmark_precision.py`, `outputs/benchmarks/precision/results.json`). Person quality is unchanged. Back to back, both take about 17 ms per 320×240 frame. At the application's rate of 5 frames/s, where the GPU drops to its P8 power state between frames (sampled clocks 210–405 MHz), FP16 lowers the median from 66.5 to 53.2 ms and the p90 from 78.6 to 60.8 ms. On the event clip, FP32 and FP16 gave the same class counts. FP16 is therefore the default, with FP32 as automatic fall-back.

Table 3. RF-DETR Nano, FP32 vs FP16 (RTX 4050 Laptop GPU, 320×240 frames, 100 timed frames each).

Precision	AP@0.5	AP@0.5:0.95	Precision @0.40	Recall @0.40	Back-to-back median / p90 ms	At 5 frames/s median / p90 ms
FP32	0.660	0.304	0.588	0.729	17.3 / 21.7	66.5 / 78.6
FP16 (used)	0.660	0.305	0.595	0.729	16.9 / 20.4	53.2 / 60.8

7.3 Serving efficiency at a fixed 5 frames/s

CPU threads. With RF-DETR on the GPU, PyTorch's default CPU thread pool (8 threads) kept the server at 712 % of one CPU core while streaming 5 frames/s, although the GPU did the detection. Two threads cut this to 121 % with no loss of frame rate (Figure 5). Median frame time was 30, 38 and 25 ms for 8, 4 and 2 threads.

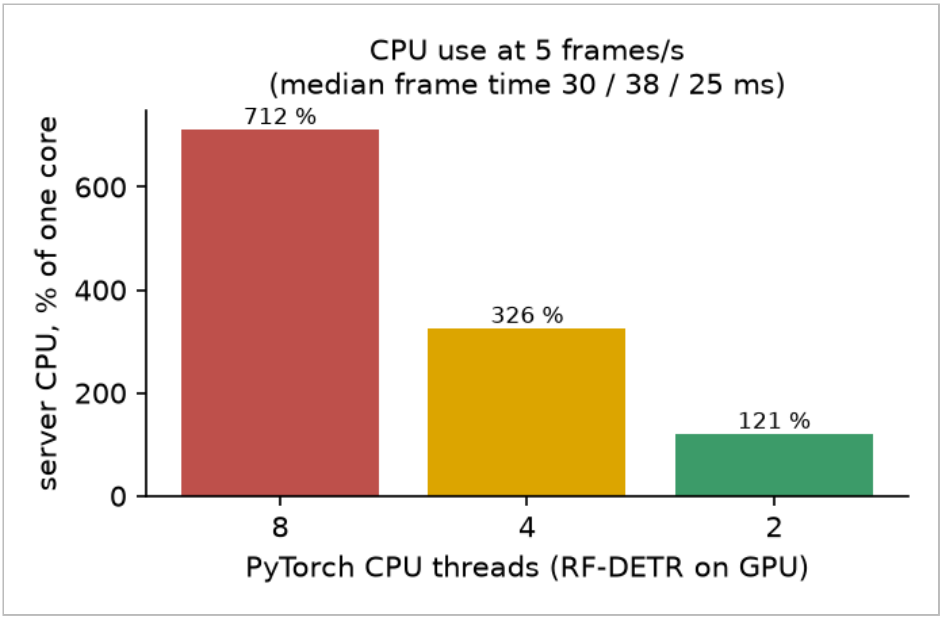


Figure 5. Server CPU use with RF-DETR on the GPU, by PyTorch CPU thread count.

GPU power state. At 5 frames/s the GPU is idle about 90 % of the time. In repeated in-process timings, the RTX 4050 then dropped to its lowest performance state (P8, about 390 MHz). The same RF-DETR inference took about 16 ms at P5 (≈600 MHz) and about 74 ms at P8. This explains why per-frame latency in the live pipeline is several times the back-to-back benchmark. FP16 reduces the effect (section 7.2). Locking clocks needs administrator rights and was not done. The Windows/NVIDIA "prefer maximum performance" setting is recommended for demonstrations.

Inference thread. The HTTP server handles each request on a new thread. Running GPU inference from a fresh thread each time cost about 3 ms per call in isolation (19.4 vs 16.2 ms median). We moved all inference to one long-lived worker thread.

7.4 Ten-minute soak tests

The server was run with a mock cloud provider that takes 20 s per description. `tools/soak_test.py` streamed the clip to the frame endpoint at 5 frames/s, one request at a time, as the phone does. Every 60 s it sent a cloud request in parallel, and it recorded 10 s windows.

Table 4. Ten-minute soak tests (`outputs/benchmarks/soak/`), 59 windows of 10 s each. "Before" ran the earlier code (FP32, inference on the HTTP request threads); "after" ran the final code (FP16, one inference thread).

Measure	Before	After
Frames sent / errors	2980 / 0	2980 / 0
Mean frames/s	4.97	4.97
Request latency: range of window medians	62–104 ms	70–78 ms

Request latency: median of window medians	92.7 ms	74.8 ms
Request latency: highest window p90	111.3 ms	92.4 ms
Server RAM (resident), min–max	851–2426 MiB	2255–2431 MiB
Server CPU, median, % of one core	142	130
GPU memory, all processes (first = last)	735 MiB	845 MiB
Parallel 20 s cloud requests answered	7 / 7	7 / 7
Events spoken	0*	148

* The first run played the clip frame by frame, so the cyclist moved at half speed (0.10 widths/s), below the 0.12 threshold. The second run sampled the clip by elapsed time, as a camera does. The event counts therefore are not comparable, and the per-frame work differed slightly between runs.

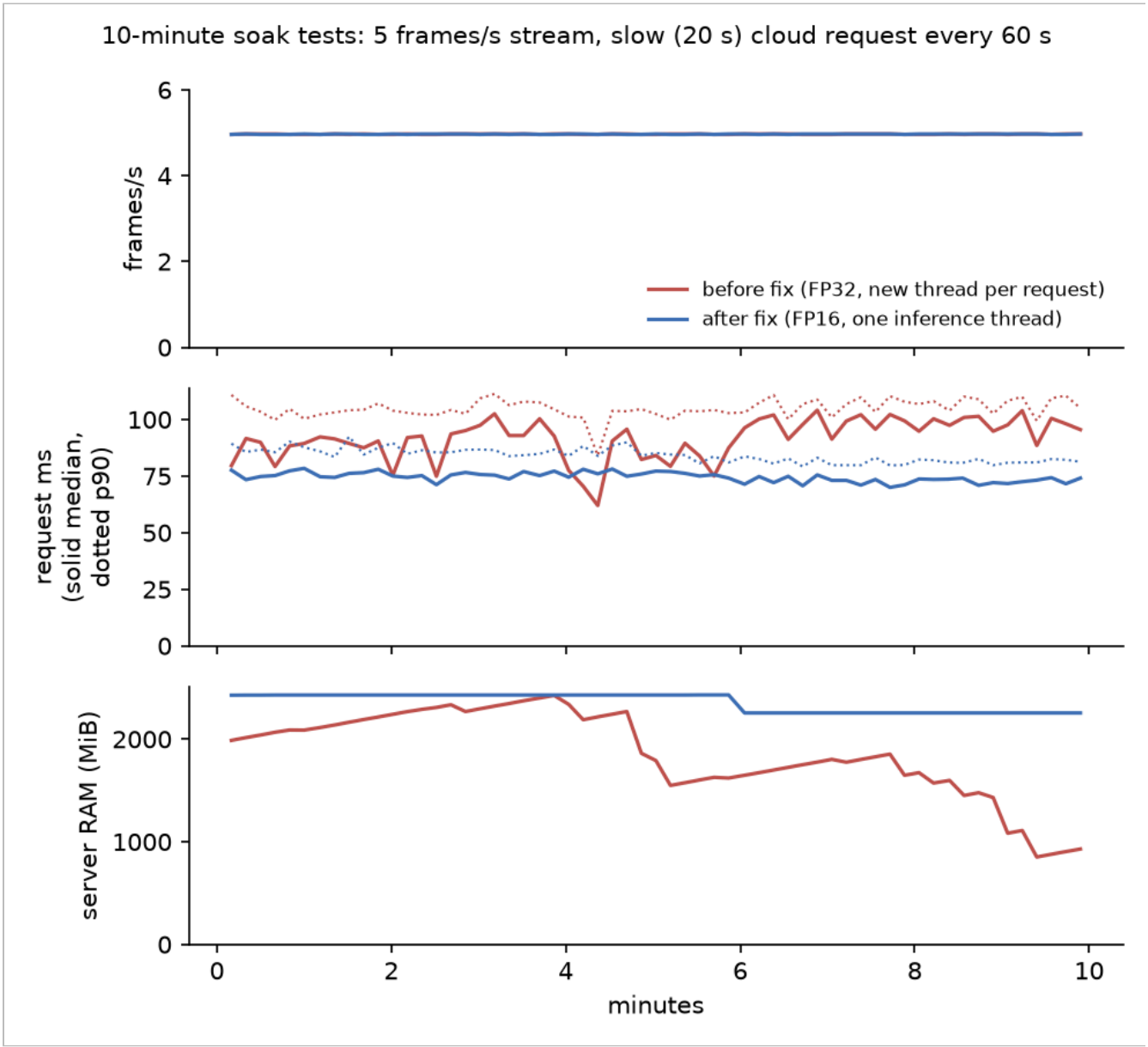


Figure 6. Soak tests before and after the serving changes.

Both runs were stable: no errors, a constant 4.97 frames/s, every parallel slow cloud request answered, and no growth in GPU memory. Server memory showed no upward trend in either run. The first run swung between 851 and 2426 MiB as memory was allocated and released; the second stayed between 2255 and 2431 MiB. After the changes, request latency was steadier (window medians 70–78 ms instead of 62–104 ms) and its worst p90 fell from 111 to 92 ms. Ten minutes on one machine cannot exclude a slow leak over hours.

7.5 End-to-end behaviour in the browser

The seven headless-Chrome tests drive the real phone interface at iPhone size against a server started from `.venv-gpu`, with the network blocked. In the event test, the server's recommended interval was 200 ms and the page achieved 4.91 frames/s. The capture-to-answer time measured in the page (JPEG encoding, HTTP on loopback, full server pipeline) had a median of 67 ms, p90 73 ms and maximum 81 ms over 44 frames. "Caution. A person is moving into your path from the left." was delivered and the neural-voice audio started 257 ms later. In the slow-cloud test, the phone said "Still waiting for the online description. Local walking guidance continues." at 15.2 s and spoke the description at 20.1 s, while 99 frames were analysed and walking-path messages continued in the meantime. The clip's subject is a person riding a bicycle, so RF-DETR reports both a person and a bicycle. In the soak test, 100 of the 148 events named the person and 48 the bicycle.

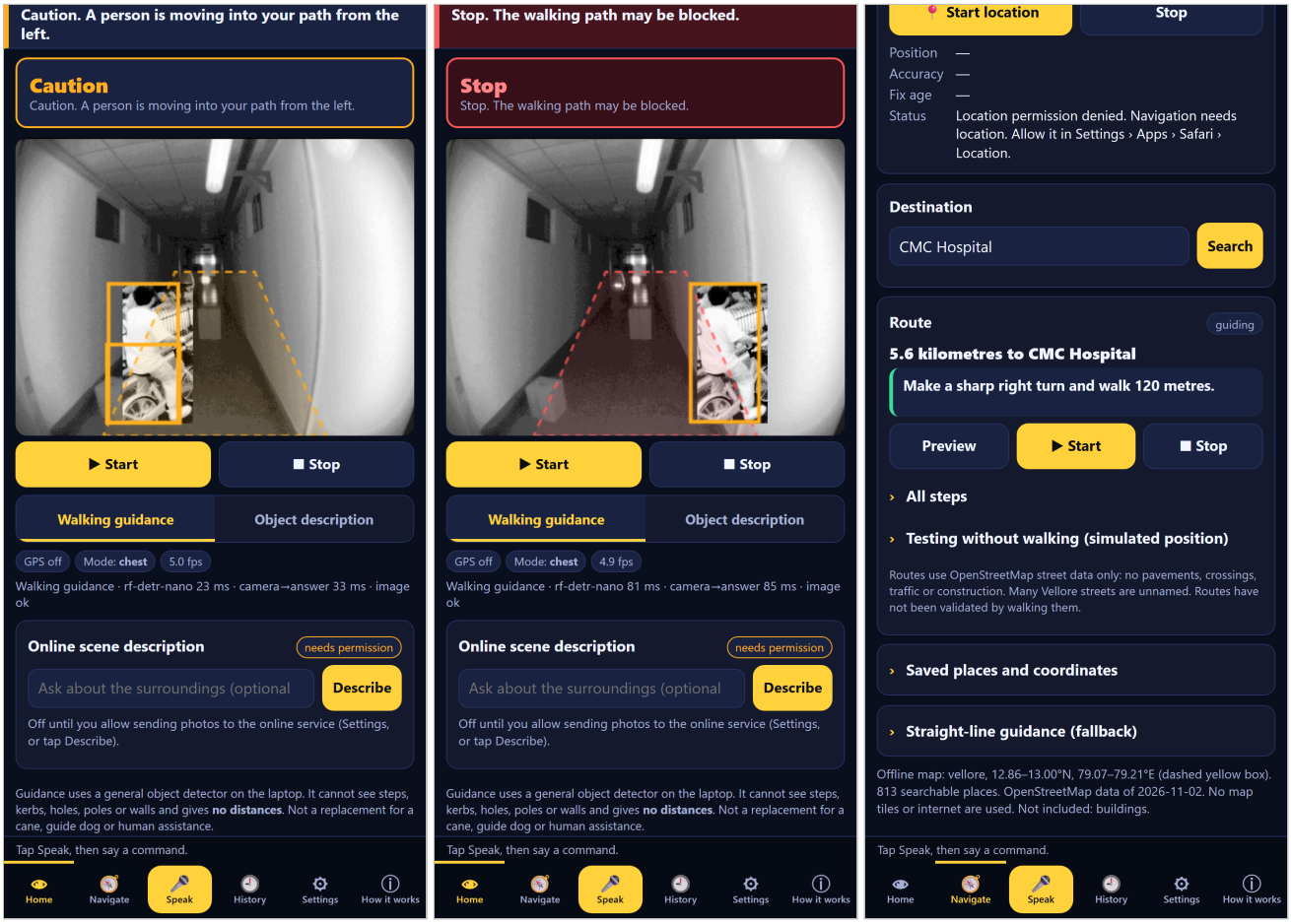


Figure 7. Phone interface in Chrome's iPhone-sized emulation with the fake camera. Left: entering event. Centre: blocked path. Right: offline navigation with a simulated position. The camera images are a composite of recorded footage, not field footage.

7.6 Automated tests

`python -m unittest discover` runs 204 test cases: unit, integration and HTTP API tests, plus 32 Node tests of the browser logic. Result: **OK** in `.venv-gpu` (1 skipped: an opt-in live cloud test) and OK in the CPU-only `.venv` (3 skipped: two GPU-only RF-DETR tests and the live test, with the app falling back to SSDLite automatically). The perception tests (22) cover:

- start-up and run-time detector fall-back;
- recovery of a known camera shift and zoom;
- stable track identities while the camera pans;
- entering and crossing in both directions;
- *no* event for a static car during a pan, for a shaky camera, in BLOCKED_VIEW, for slow drift and for chairs;
- the absence of distance and safety words.

All 7 browser tests pass against the final code: smoke, voice and map, refined interface, final integration, cloud description with a simulated outage, slow cloud description, and events. A quality-gate launch with the documented command also passed: frames, speech, place search, a 1330 m route with 10 steps, and a voice command, from both the GPU environment (RF-DETR, FP16, 200 ms) and the CPU environment (SSDLite fall-back, 400 ms).

8. Discussion

Objective O1 (frame rate and latency). Met on the laptop. The pipeline sustained ≈5 frames/s with a capture-to-answer median well below the 200 ms interval. The decisive factors were serving details, not the detector's benchmark speed:

- idle CPU threads;
- per-request threads;
- above all, GPU clock scaling at low load.

A detector that takes 21 ms back to back took several times longer at 5 frames/s. Benchmarks for assistive systems should therefore be run at the frame rate of the application, not only back to back.

Objective O2 (detector choice). Met, within the limits of the data. RF-DETR Nano was the most accurate of the four on the only annotated person data available. Its lower CPU speed is covered by the SSDLite fall-back. Whether the ranking holds for phone RGB images of Indian streets is untested.

Objective O3 (motion without false alarms from walking). Partly shown. In tests with a synthetically panning camera, a static car gave no event, while a person moving into the path did. The price is conservatism:

- events need about 1 s of confirmation;
- events are suppressed while the camera sways strongly, which can include normal walking with a hand-held phone.

Missed events are therefore expected. This is acceptable only because the system is supplementary to the cane.

Objectives O4–O6 (offline, wording, verification). Met in automated tests:

- offline operation was tested by blocking the network in-process and in Chrome;
- a safety filter removes distances and "safe"-type words from cloud descriptions;
- the test suites pass in both environments.

Physical verification is the main gap (section 9).

9. Limitations and threats to validity

- **No iPhone or field validation.** RF-DETR has not yet processed real iPhone frames. Wi-Fi latency, outdoor lighting, and use by visually impaired people are untested. The protocol in `docs/IPHONE_VALIDATION_PROTOCOL.md` lists 23 scenarios for these tests.
- **Proxy data.** Detector quality was measured on near-infrared images with 129 people. The live tests use a fake camera and a composited clip.
- **Scope of perception.** 23 COCO classes only. No steps, kerbs, holes, poles, walls or wet floors. No distance, and no "approaching" alerts.
- **Dependence.** The phone needs the laptop and local Wi-Fi. iOS Safari stops the camera when the screen locks. The GPU path needs an NVIDIA GPU.
- **Single machine.** All timings come from one laptop and one GPU driver, and power-state behaviour differs between machines.
- **Cloud description.** The single approved live Gemini request timed out. Cloud quality and latency are unmeasured.
- **Licences.** Piper is GPL-3.0+ and the voice model AGPL-3.0, which constrains redistribution.

10. Ethics and safety

NeuroSonic is a research prototype. It is **not** a mobility aid and must not replace a white cane, a guide dog or human assistance. Its messages say "appears clear", never "safe". It never estimates distance and never claims 360-degree awareness.

Camera images are processed in memory and are not stored. The cloud description sends one image to a third party only after the user opts in.

Any field test must follow the safety rules in the validation protocol: a sighted companion, a cane in use, controlled routes, and stopping on any doubt.

11. Conclusion and future work

A consumer laptop GPU can give a browser-based phone client real-time, offline, spoken object and movement awareness with a modern detector. In our tests it sustained about 5 frames/s and answered within roughly 50–100 ms on the laptop.

The engineering lessons are general:

- measure at the application's frame rate;

- constrain CPU threads when the GPU does the work;
- keep GPU work on one thread.

The next step is not a new model but evidence: the iPhone and field protocol. After that, the planned work is:

- monocular depth to support "approaching" alerts;
- ground-hazard data (kerbs, steps, potholes) under open licences;
- a native application with on-device inference that removes the laptop dependency.

Data and code availability

All code, test scripts, benchmark outputs (`outputs/benchmarks/`) and the figure generator (`tools/make_publication_figures.py`) are in the project directory. The commands to reproduce every number are in `publication/TESTING_AND_VALIDATION_REPORT.md`.

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